



2025–2026 Undergraduate Catalog

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Physics/Engineering Dual Degree Program (B.S.)



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120 hours

The University of North Georgia offers a dual degree program which allows students to receive a bachelor's degree in physics from UNG and a bachelor's degree in Engineering from a participating university. Students are provided the opportunity to attend the University of North Georgia for approximately three years before embarking on approximately two years of engineering coursework at the University of Georgia, Georgia Institute of Technology, Georgia Southern University, Kennesaw State University, Clemson University, or Mercer University. After completing the academic requirements of both institutions, the student receives a bachelor's degree in physics from the University of North Georgia and a bachelor's degree in a chosen engineering field from the second institution.

UNG faculty and staff work closely with the faculty of the engineering schools to assure a curriculum which is well-coordinated. Dual degree students first satisfactorily complete the UNG required curriculum and pre-engineering curriculum required by the transfer institution, and then apply for transfer to the engineering school. However, admission to the engineering

school will also be based upon the specific requirements of each program, such as overall grade point average, math–science grade point average, and other factors determined by that institution. These requirements vary, and each school’s transfer expectations and requirements are listed on their website. UNG faculty and staff advisors in the related dual degree field should also be consulted for their assistance with the dual degree process.

Dual Degree Requirements:

1. Complete the course work required by UNG as given below.
2. Complete the course work required to transfer to the chosen institution with their required GPA.
3. Complete the required engineering program at the transfer institution.
4. Apply for the second B.S. degree from UNG.

Core IMPACTS – 42 hours

Refer to Core IMPACTS

Institution: No exceptions

Mathematics: MATH 1113 or MATH 1450

Citizenship: No exceptions

Humanities: No exceptions

Writing: No exceptions

STEM: CHEM 1211/1211L and CHEM 1212/1212L and MATH 1450 or MATH 2460 (if MATH 1450 was taken in IMPACTS Mathematics) recommended

Social Sciences: No exceptions

Field of Study – 18 hours

Required Courses

- PHYS 2211 – Principles of Physics I
- PHYS 2211L – Principles of Physics I Lab
- (or PHYS 2211K – Principles of Physics I)
- PHYS 2212 – Principles of Physics II
- PHYS 2212L – Principles of Physics II Lab
- (or PHYS 2212K – Principles of Physics II)
- CSCI 1371 – Computing for Scientists and Engineers

For Students with MATH 1113 in Mathematics

- MATH 2460 – Calculus II
- MATH 2470 – Calculus III

For Students with MATH 1450 in Mathematics

- MATH 2470 – Calculus III
- 2 carryover hours from Core IMPACTS Mathematics and Core IMPACTS STEM
- 1 elective hour in any 1–2000 level ENGR or Core IMPACTS STEM course (except MATH 1113 or CSCI 1250)

Major Requirements – 60 hours

Required Courses

- PHYS 3111 – Mechanics I
- PHYS 3310 – Modern Physics
- PHYS 3310L – Modern Physics Laboratory
- PHYS 3411 – Electricity and Magnetism I
- PHYS 3610 – Thermal Physics

CHEM 1211/1211L and CHEM 1212/1212L must be taken here if not taken in IMPACTS STEM.

- CHEM 1211 – Principles of Chemistry I
- CHEM 1211L – Principles of Chemistry I Laboratory
- CHEM 1212 – Principles of Chemistry II
- CHEM 1212L – Principles of Chemistry II Laboratory

Electives

- To bring the total number of University of North Georgia credit hours to 90 and total credit hours to 120 for the degree.

Other Requirements

- MATH 3000 – Differential Equations
- US and Georgia History Requirements
- US and Georgia Constitution Requirements
- Regents' Reading and Writing Requirements

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